David Alexander Tedjopurnomo

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Education

RMIT University, PhD in Computer Science **2023**

- Received the CSIRO Data 61 scholarship and scholarship top-up
- Received the RMIT 2020 Research Achievement Award

The University of Melbourne, Master of Information Technology **2017**

The University of Newcastle, Bachelor of Information Technology **2014**

Experience

RMIT University, Lecturer (Education Focused) (ACDF) **06.2025 - Present**

- Delivering lectures and participating in the course coordination and development for the first semester Bootcamp and Studio courses.

Swinburne University of Technology, Postdoctoral Research Fellow **05.2024 - 12.2024**

- Participated in an Australian Research Council-funded research on traffic incident management
- Performed traffic incident simulations using Aimsun and Eclipse SUMO
- Mined, cleaned and integrated traffic flow and traffic incidents data from Sydney and Sacramento
- Developed machine learning models for future traffic flow prediction

RMIT University, University Tutor **02.2023 - 06.2025**

- Tutored the following courses: Java Programming Bootcamp 1, Java Programming Studio 1, Python Programming Bootcamp 1, Python Programming Studio 1, Web Programming Bootcamp, Web Programming Studio, Performance Management Foundations
- Achieved a 4.02 out of 5.00 Good Teaching Score (GTS). Exceeding the school, college and RMIT mean GTS score in 2023

The University of Melbourne, University Tutor **02.2023 – 05.2024**

- Tutored the following courses: Advanced Database Systems, Geospatial Data Management, Research Project

Selected Publications

David Alexander Tedjopurnomo, Zhifeng Bao, Farhana Choudhury, Hui Luo, and A. K. Qin. 2022. Equitable Public Bus Network Optimization for Social Good: A Case Study of Singapore. In *2022 ACM Conference on Fairness, Accountability, and Transparency (FAccT '22)*. Association for Computing Machinery, New York, NY, USA, 278–288. https://doi.org/10.1145/3531146.3533092

David Alexander Tedjopurnomo, Zhifeng Bao, Baihua Zheng, Farhana Choudhury and A. K. Qin. 2022. A Survey on Modern Deep Neural Network for Traffic Prediction: Trends, Methods and Challenges. In *IEEE*

Transactions on Knowledge and Data Engineering, vol. 34, no. 4, pp. 1544-1561. https://doi.org/10.1109/TKDE.2020.3001195

David Alexander Tedjopurnomo, Xiucheng Li, Zhifeng Bao, Gao Cong, Farhana Choudhury, and A. K. Qin. 2021. Similar Trajectory Search with Spatio-Temporal Deep Representation Learning. In *ACM Transactions on Intelligent Systems and* Technology, vol. 12, no. 6, Article 77 (December 2021), pp 1-26. https://doi.org/10.1145/3466687

Other Research Experiences and Awards

Research awards: CIKM 2024 Distinguished Reviewer Award

Program committee member: CIKM 2024, CIKM 2025, WWW 2026

Conference Reviewing: VLDB 2023, ADC 2023-2024

Journal Reviewing: T-ITS 2023-2025, TOIS 2023-2024, IEEE Access 2023-2025, DSE 2024-2025, ITSM 2024-2025, IET Networks 2024, Pattern Analysis and Applications 2024, Applied Artificial Intelligence 2023-2024, IEEE IoT 2026